

Volume of prisms



- 1
- | | | | | | | | |
|---|----------------------|---|---------------------|---|------------------------|---|----------------------|
| a | 270 cm^3 | b | 48 cm^3 | c | 32 cm^3 | d | 217.5 cm^3 |
| e | 330 cm^3 | f | 68 cm^3 | g | 7164 units^3 | h | 1080 cm^3 |
| i | 187.7 cm^3 | j | 84 cm^3 | k | 280 cm^3 | l | 828 cm^3 |
| m | 399 cm^3 | n | 412.5 m^3 | | | | |

- 2
- | | | | |
|---|----------------------|---|------------------------|
| a | 32.5542 cm | b | $18\,064 \text{ cm}^3$ |
|---|----------------------|---|------------------------|

- 3
- 47 mm^3

- 4
- 300 cm^3

- 5
- | | | | | | |
|---|-----------------------|---|-----------------------|---|------------------------|
| a | 1198.5 cm^3 | b | 299.625 m^3 | c | $h = 0.5875 \text{ m}$ |
|---|-----------------------|---|-----------------------|---|------------------------|

- 6
- 1834.00 m^3

Volume of pyramids

- 7
- | | | | | | | | |
|---|-----------------------|---|--------------------|---|--------------------|---|--------------------|
| a | 154.78 cm^3 | b | 208 m^3 | c | 80 cm^3 | d | 864 cm^3 |
| e | 16 cm^3 | f | 200 cm^3 | g | 160 cm^3 | | |

- 8
- 240 cm^3

- 9
- $h = 12 \text{ cm}$

- 10
- | | | | | | |
|---|-------------------------|---|------------------------|---|-------------------------|
| a | $RT = 38.12 \text{ cm}$ | b | 1677.28 cm^2 | c | 38018.35 cm^3 |
|---|-------------------------|---|------------------------|---|-------------------------|

- 11
- 12 cm

- 12
- $\frac{448}{3} \text{ cm}^3$



Volume of cylinders

- 13 a 785.4 cm^3 b 307.9 cm^3 c 70.7 cm^3 d 1608.5 cubic units
 e 883.6 cm^3 f 491.5 cm^3 g 609.5 cm^3

14 2309.07 cm^3

15 3.62 Litres

Volume of cones

- 16 a 25.13 cm^3 b 32.45 cm^3

- 17 a $h = 24 \text{ m}$ b 2510 m^3

18 452.4 cm^3

19 $r = 5.72 \text{ mm}$

20 43.37 m^3

Volume of a sphere

- 21 a 113.10 cm^3 b 4577.20 cm^3 c $113\,550.33 \text{ mm}^3$ d 33.51 cm^3

- 22 a $288\pi \text{ cm}^3$ b $972\pi \text{ cm}^3$ c $288\pi \text{ m}^3$ d $\frac{32\pi}{3} \text{ cm}^3$

23 The volume is multiplied by 8

- 24 a $r = 3.42 \text{ cm}$ b $r = 4.41 \text{ cm}$

25 549.78 cm^3



26

a 46173.484 cm^3

b 46 L

Applications

27

a 982.35 cm^3

b 169.65 cm^3

c 292.17 cm^3

d 469.14 cm^3

e 221.74 cm^3

f 263.89 cm^3

g 735.13 cm^3

h 720.00 cm^3

i 222.83 m^3

28

a $h = 15 \text{ cm}$

b $V = 916.72 \text{ cm}^3$

29

a 29.69 cm^3

b 169.2 g

30 0.57 m^3

31

a 70.55 km^3

b 0.15 km^3

c 70.40 km^3

32

a 23 m^3

b 14 137 Litres

33

a 1335.18 cm^3

b 4005.54 cm^3

c 28 L

34

a 3

b 6158 cm^3

c $18\,473 \text{ cm}^3$

d Large container

35

a 804.2 cm^3

b 1206.4 cm^3

c 33%